Assessment: Observational Constraints on Two-Parameter Dark-Energy Parametrizations

Context and purpose

This assessment extends the flat ΛCDM inference pipeline developed in the previous weeks to three phenomenological two-parameter dark-energy parametrizations. Students must use the same observational framework already implemented for Type Ia supernovae, Cosmic Chronometers, and BAO, and replace only the background expansion function $E(z)$ according to each dark-energy model.

The analysis must be performed in a spatially flat Friedmann–Lemaître–Robertson–Walker background. Radiation may be neglected for the late-time data used in this exercise. The purpose of the assessment is not only to obtain numerical constraints, but also to evaluate the physical viability of each parametrization and its possible cosmological implications.

Datasets

Students must use the following three observational probes:

(i) Type Ia supernovae from the Pantheon+SH0ES sample, using the full covariance matrix already implemented in the previous weeks.

(ii) Cosmic Chronometers, treated as direct measurements of $H(z)$ with observational uncertainties.

(iii) BAO measurements, using the appropriate observable for each data point, e.g. $D_M(z)/r_d$, $D_H(z)/r_d$, $D_V(z)/r_d$, or $H(z)r_d$, depending on the dataset provided.

Each dataset must first be analysed independently. A final joint analysis must then be performed using the combined likelihood

$$\ln \mathcal{L}_{\rm total} = \ln \mathcal{L}_{\rm SN} + \ln \mathcal{L}_{\rm CC} + \ln \mathcal{L}_{\rm BAO}, \tag{1}$$

assuming statistical independence among the three datasets.

Cosmological framework

For all models, the Hubble expansion rate is written as

$$H(z) = H_0 E(z), \tag{2}$$

where H_0 is the Hubble constant and $E(z)$ is the dimensionless expansion function.

The parameter vector for the full joint analysis must be

$$\boldsymbol{\theta} = (\Omega_m, H_0, M_B, r_d, w_0, w_a), \tag{3}$$

or

$$\boldsymbol{\theta} = (\Omega_m, H_0, M_B, r_d, w_0, w_1), \tag{4}$$

for the Barboza–Alcaniz case. Here M_B is the absolute-magnitude nuisance parameter entering the supernova likelihood, and r_d is the sound horizon at the baryon drag epoch entering the BAO likelihood.

The flatness condition implies

$$\Omega_{\rm DE} = 1 - \Omega_m. \tag{5}$$

Dark-energy parametrizations

Students must analyse the following three two-parameter parametrizations.

1. Chevallier–Polarski–Linder parametrization

The CPL parametrization is given by

$$w(z) = w_0 + w_a \frac{z}{1+z}, \tag{6}$$

or equivalently $w(a) = w_0 + w_a(1-a)$. The corresponding expansion function is

$$E^2_{\rm CPL}(z) = \Omega_m(1+z)^3 + (1-\Omega_m)(1+z)^{3(1+w_0+w_a)} \exp\left[-\frac{3w_a z}{1+z}\right]. \tag{7}$$

The ΛCDM limit is recovered for

$$w_0 = -1, \qquad w_a = 0. \tag{8}$$

2. Jassal–Bagla–Padmanabhan parametrization

The JBP parametrization is

$$w(z) = w_0 + w_a \frac{z}{(1+z)^2}. \tag{9}$$

Its dark-energy density evolves as

$$\rho_{\rm DE}(z) \propto (1+z)^{3(1+w_0)} \exp\left[\frac{3w_a z^2}{2(1+z)^2}\right], \tag{10}$$

and therefore

$$E^2_{\rm JBP}(z) = \Omega_m(1+z)^3 + (1-\Omega_m)(1+z)^{3(1+w_0)} \exp\left[\frac{3w_a z^2}{2(1+z)^2}\right]. \tag{11}$$

The ΛCDM limit is again

$$w_0 = -1, \qquad w_a = 0. \tag{12}$$

3. Barboza–Alcaniz parametrization

The Barboza–Alcaniz parametrization is

$$w(z) = w_0 + w_1 \frac{z(1+z)}{1+z^2}. \tag{13}$$

The dark-energy density evolves as

$$\rho_{\rm DE}(z) \propto (1+z)^{3(1+w_0)}(1+z^2)^{3w_1/2}, \tag{14}$$

leading to

$$E^2_{\rm BA}(z) = \Omega_m(1+z)^3 + (1-\Omega_m)(1+z)^{3(1+w_0)}(1+z^2)^{3w_1/2}. \tag{15}$$

The ΛCDM limit is

$$w_0 = -1, \qquad w_1 = 0. \tag{16}$$

Observable quantities

For each parametrization, students must compute the following quantities consistently.

Supernovae

For the supernova likelihood, the theoretical apparent magnitude must be written as

$$m_{b,\rm th} = M_B + \mu_{\rm th}(z), \tag{17}$$

where

$$\mu_{\rm th}(z) = 5\log_{10}\left[\frac{D_L(z)}{\rm Mpc}\right] + 25. \tag{18}$$

For a spatially flat model,

$$D_L(z) = (1+z)D_M(z), \tag{19}$$

with

$$D_M(z) = c\int_0^z \frac{dz'}{H(z')}. \tag{20}$$

If the Pantheon+SH0ES implementation distinguishes between heliocentric and Hubble-diagram redshifts, students must preserve the same convention used in the previous weeks.

Cosmic Chronometers

For Cosmic Chronometers, the theoretical prediction is simply

$$H_{\rm th}(z) = H_0 E(z). \tag{21}$$

The Gaussian likelihood may be written as

$$-2\ln\mathcal{L}_{\rm CC} = \sum_i \left[\frac{H_{\rm obs}(z_i) - H_{\rm th}(z_i)}{\sigma_{H,i}}\right]^2 + \sum_i \ln(2\pi\sigma^2_{H,i}). \tag{22}$$

BAO

For BAO, students must compute the appropriate observable for each data point. The relevant distance quantities are

$$D_H(z) = \frac{c}{H(z)}, \tag{23}$$

$$D_M(z) = c\int_0^z \frac{dz'}{H(z')}, \tag{24}$$

and

$$D_V(z) = \left[z D_H(z) D_M^2(z)\right]^{1/3}. \tag{25}$$

The BAO likelihood must use the correct observable labels and must not treat all BAO measurements as if they were the same quantity.

If a covariance matrix is provided, the BAO contribution must be computed as

$$\chi^2_{\rm BAO} = \Delta^T_{\rm BAO} C^{-1}_{\rm BAO} \Delta_{\rm BAO}, \tag{26}$$

where the ordering of the residual vector must match the ordering of the covariance matrix exactly.

Required analysis

For each of the three parametrizations, students must perform the following analyses:

A1. Supernova-only inference.

A2. Cosmic-Chronometer-only inference.

A3. BAO-only inference.

A4. Joint inference using SN+CC+BAO.

Each run must use a clearly stated prior. The same prior ranges should be used across the three parametrizations unless a physically motivated exception is justified.

A possible baseline choice is

$$0.01 < \Omega_m < 0.6, \tag{27}$$

$$50 < H_0 < 90, \tag{28}$$

$$-20.5 < M_B < -18.0, \tag{29}$$

$$120 < r_d/\mathrm{Mpc} < 180, \tag{30}$$

$$-2.5 < w_0 < -0.3, \tag{31}$$

$$-3.0 < w_a < 3.0, \tag{32}$$

with w_a replaced by w_1 for the Barboza–Alcaniz model. Students may adopt different priors only if they explicitly justify the choice.

Statistical requirements

Students must perform Bayesian inference using MCMC or an equivalent posterior-sampling method. For each model and dataset combination, they must report:

(i) the posterior median of each free parameter;

(ii) the 68% credible interval, interpreted as the 1σ region;

(iii) the 95% credible interval, interpreted approximately as the 2σ region;

(iv) the best-fit or maximum-posterior parameter values;

(v) the minimum χ^2 or maximum log-likelihood;

(vi) convergence diagnostics.

For two-dimensional parameter planes, students must produce confidence contours at 1σ and 2σ for at least:

$$(w_0, w_a), \qquad (H_0, \Omega_m), \qquad (H_0, r_d), \tag{33}$$

where applicable. For Barboza–Alcaniz, (w_0, w_a) must be replaced by (w_0, w_1).

Required figures

The final report must include at least the following figures:

F1. For each dark-energy parametrization, students must produce separate corner plots for each individual dataset and for the combined analysis:

(a) SN-only posterior,
(b) CC-only posterior,
(c) BAO-only posterior,
(d) joint SN+CC+BAO posterior.

If a dataset alone leaves some parameters weakly constrained, the corresponding corner plot must still be shown and discussed rather than omitted.

F2. One corner plot for each parametrization using the joint SN+CC+BAO posterior.

F3. One comparison plot of the 1σ and 2σ contours in the dark-energy parameter plane for CPL, JBP, and Barboza–Alcaniz.

F4. One plot of $H(z)$ showing the CC data and the posterior prediction for each model.

F5. One supernova Hubble diagram or residual plot for the joint best-fit solution.

F6. One BAO residual plot, using normalized residuals $(d_i - t_i)/\sigma_i$ or the equivalent covariance-weighted diagnostic.

F7. One plot of $w(z)$ over the redshift range covered by the data, showing the posterior median and the 68% credible band.

Figures must have labelled axes, units where appropriate, readable legends, and captions that explain what is being compared.

Model comparison and physical interpretation

Students must discuss whether each parametrization is observationally viable. The discussion must address the following points:

Q1. Is the ΛCDM point contained within the 1σ or 2σ region of the dark-energy parameter plane?

Q2. Does the model prefer quintessence-like behaviour, phantom-like behaviour, or a crossing of the phantom divide $w = -1$?

Q3. How do the constraints change when moving from individual datasets to the joint SN+CC+BAO analysis?

Q4. Which dataset is most sensitive to H_0? Which one most strongly constrains the geometrical combination involving r_d?

Q5. Are the inferred values of r_d physically reasonable when compared with standard expectations, without imposing an external early-Universe prior?

Q6. Does the model show signs of overfitting relative to flat ΛCDM?

Q7. What are the possible cosmological implications if a model favours $w(z) \neq -1$ at more than 2σ?

Students must also compute information criteria for the joint analysis:

$$\mathrm{AIC} = -2 \ln \mathcal{L}_{\max} + 2k, \tag{34}$$

$$\mathrm{BIC} = -2 \ln \mathcal{L}_{\max} + k \ln N, \tag{35}$$

where k is the number of free parameters and N is the total number of data points. These criteria must be compared against the flat ΛCDM baseline fitted to the same data combination.

Deliverables

Students must submit:

D1. A short scientific report of 6–10 pages.

D2. The complete Python code used in the analysis.

D3. A README file explaining how to reproduce the results.

D4. Posterior sample files for all joint runs.

D5. Tables of parameter constraints for all individual and joint analyses.

D6. All figures in PDF or PNG format.

The report must be written in the style of a compact observational cosmology paper, with sections for methodology, data, results, discussion, and conclusions.

Suggested structure of the report

1. Introduction and motivation.
2. Datasets and likelihood construction.
3. Dark-energy parametrizations.
4. Numerical method and priors.
5. Results from individual datasets.
6. Results from the joint SN+CC+BAO analysis.
7. Model comparison with flat ΛCDM.
8. Cosmological interpretation and conclusions.

Assessment rubric

Criterion	**Description**	**Weight**
Data handling	Correct reading, validation, unit treatment, and covariance use for SN, CC, and BAO.	15%
Model implementation	Correct implementation of CPL, JBP, and Barboza–Alcaniz background functions.	15%
Likelihood construction	Correct individual and joint likelihoods, including covariance treatment where required.	15%
MCMC and convergence	Appropriate priors, sampling strategy, burn-in treatment, thinning where justified, and convergence diagnostics.	15%
Parameter constraints	Correct extraction and presentation of medians, 1σ intervals, 2σ intervals, and confidence contours.	15%
Model comparison	Correct use and interpretation of AIC/BIC relative to flat ΛCDM.	10%
Physical interpretation	Clear discussion of viability, phantom/quintessence behaviour, ΛCDM consistency, and cosmological implications.	10%
Reproducibility and presentation	Clean code, README, readable figures, and well-structured scientific report.	5%

Important methodological cautions

- Do not quote parameter constraints from chains that have not converged.
- Do not mix BAO observables with different dimensions in a single plot unless they are clearly separated or normalized.
- Do not impose external CMB or Planck priors unless explicitly stated as an additional test.

- Do not interpret AIC or BIC values unless all models are fitted to exactly the same dataset combination.
- Do not claim evidence for dynamical dark energy unless the statistical significance is quantified and the role of priors is discussed.

References

References

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[3] H. K. Jassal, J. S. Bagla, and T. Padmanabhan, “WMAP constraints on low redshift evolution of dark energy,” *Monthly Notices of the Royal Astronomical Society* **356**, L11–L16 (2005), arXiv:astro-ph/0404378.

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